

KEONWOO OH

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EDUCATION

University of Maryland Aug 2021 - Present
Ph.D. in Computer Science, *Advisor:* Amol Deshpande, *GPA:* 3.8/4.0

Swarthmore College Aug 2015 - May 2021
B.A. in Computer Science and Mathematics, *GPA:* 3.9/4.0

RESEARCH INTERESTS

Metadata management and **concurrency control mechanism** in large-scale data lakehouse systems. **Data formats** for efficient data asset storage in data and machine learning pipelines. Broader topics in **database system design**.

PUBLICATIONS

Keonwoo Oh, Pooja Nilangekar, and Amol Deshpande. *TreeCat: Standalone Catalog Engine for Large Data Systems*. Proc. VLDB Endow. 18, 11 (July 2025), 4323-4336.

PROJECTS

Raven
Implemented a file-level metadata system for data lakehouse architectures with a specialized focus on garbage collection.

TreeCat (<https://github.com/umddb/treecat>)
Implemented a unified, standalone catalog engine for large data lake systems. Published in Proc. VLDB 18, 11 (July 2025).

WORK EXPERIENCE

Dolby Laboratories May 2026 - Aug 2026
PhD Research Intern *Atlanta, GA*

- Evaluated and benchmarked tabular storage formats for multimodal data management in data platform.

Dolby Laboratories May 2025 - Aug 2025
PhD Research Intern *Atlanta, GA*

- Designed and implemented a file-level metadata system for data lakehouse architectures.

Vertica May 2022 - Aug 2022
Systems Software Engineering Intern *Cambridge, MA*

- Integrated the internal profiling tool into the HTTPS service via a REST API.

TEACHING EXPERIENCE

Teaching Assistant, CMSC416: Introduction to Parallel Computing, UMD Spring 2023, Spring 2024, Fall 2024
Teaching Assistant, CMSC714: High Performance Computing Systems, UMD Fall 2023
Teaching Assistant, CMSC424: Database Design, UMD Fall 2022
Teaching Assistant, CMSC250: Discrete Structures, UMD Spring 2022
Teaching Assistant, CMSC216: Introduction to Computer Systems, UMD Fall 2021

SKILLS

Programming Languages: C++, C, Scala, Python, Java, Rust, OCaml

Systems: PostgreSQL, MySQL, Apache Spark, Apache Iceberg, Delta Lake, DuckDB, RocksDB, gRPC